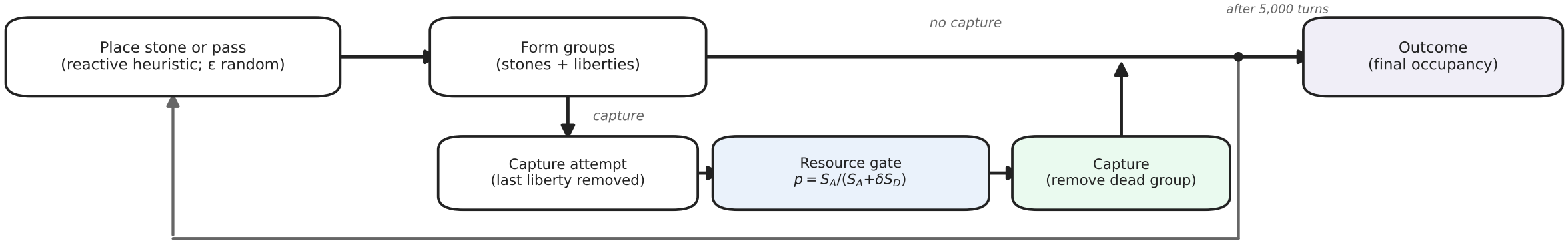
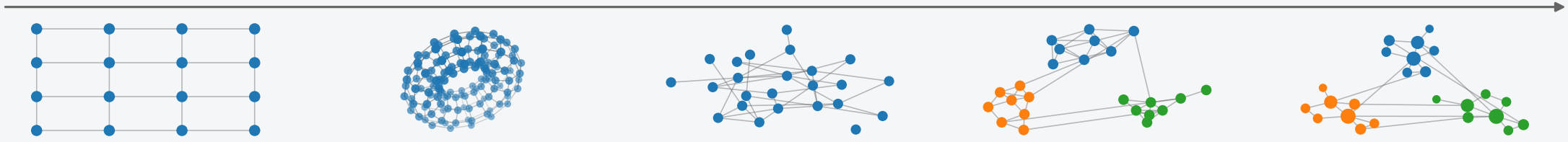
A Per-turn process



B Swept design space

Topology (regular → heterogeneous)



Competition structure

- 2-player direct
- 3-player RPS (cyclic)
- 3-player AVA (symmetric)

Scheduler

- synchronous (round-robin)
- energy-weighted asynchronous (Gillespie-like)

Other swept parameters: mean degree k ∈ {8, 10, 12, 14, 19} (lattice, torus: k ≈ 4) · board size n ∈ {81, 225, 361, 625} · exploration ε = 0.15

ε fixed at 0.15 in production, also swept {0.02, 0.04, 0.07, 0.15} for sensitivity on a reduced grid (n ∈ {225, 625}, k ∈ {10, 14}) · ≈ 50,000 simulation runs in total